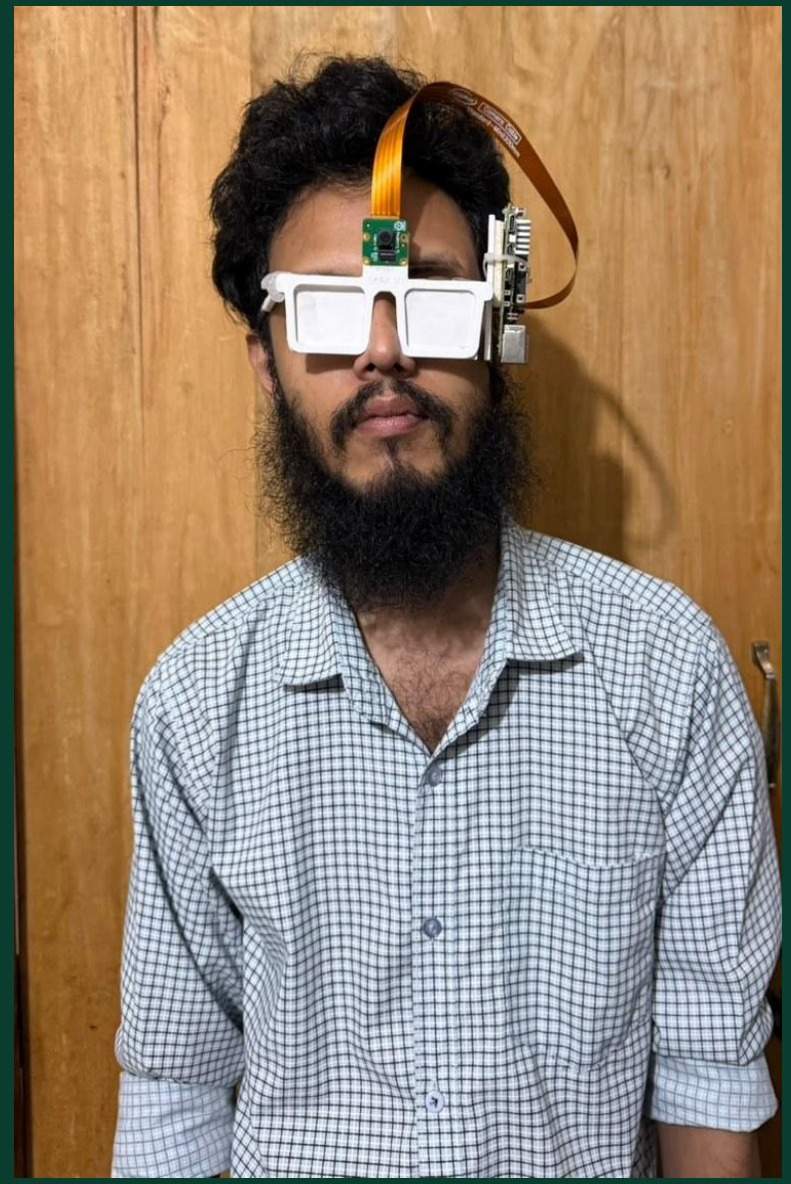
SARA

Smart Assistive Reading Aid: An Offline Smart Glass for Visually Impaired People that Reads Bangla and English Text, Detects Objects, and Recognises and Checks Bangladeshi Taka Notes

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Department of Computer Science and Engineering, Independent University, Bangladesh · Spring 2026



Reading, recognising and paying still need a helper

A blind person buys medicine and receives change. To know what the label says, which note came back and whether it is real, they must ask someone, and they cannot check the answer.

2.2 billion

people live with some form of vision impairment (WHO)



Built for English

Existing devices do not read Bangla or know Taka notes.



Need internet or money

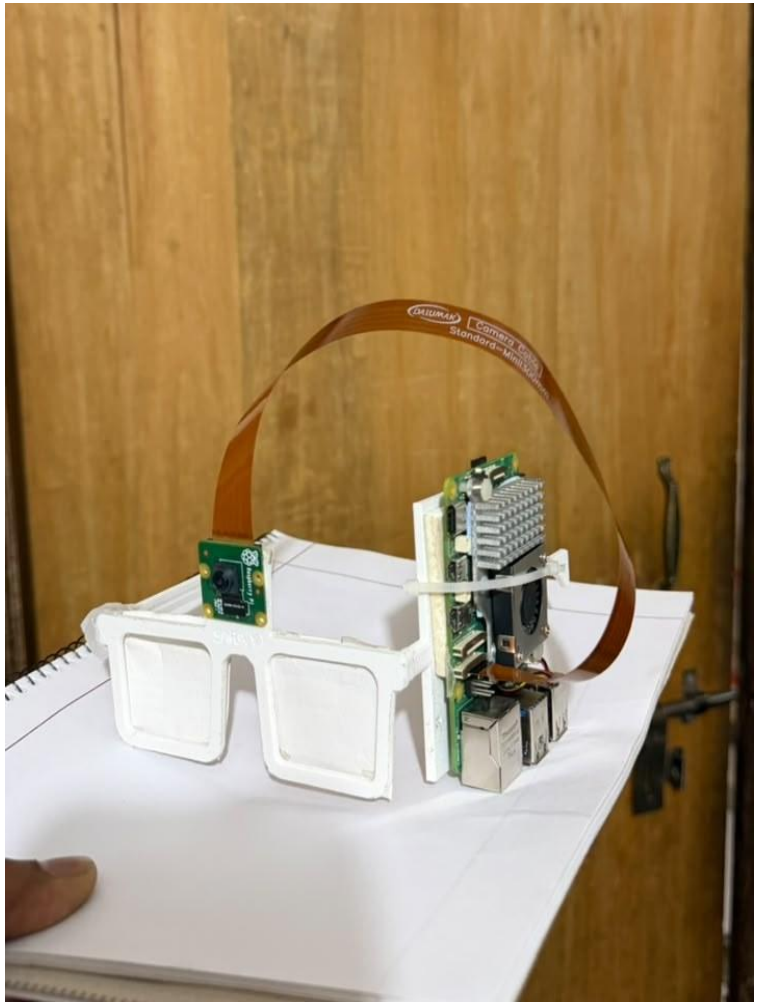
Many send pictures to a server or need a phone and a paid service.



Answers cannot be checked

A wrong note or a false "counterfeit" costs money or causes an accusation.

SARA: five modes, five buttons, Bangla speech



Text reading

Bangla and English OCR, spoken right after capture

Objects

Names objects in front of the user every 1.5 s

Currency

All nine Taka notes, with a cautious counterfeit check

Emotion

Facial expression of the person in front

Online description

Optional; the only mode that needs the internet

Controlled by MODE, ACTION, READ and two volume buttons. Every change is spoken.

Six research questions

1

System

Can a multi-function Bangla glass run on a Raspberry Pi 5, and how fast?

2

Synthetic data

Can a Taka detector trained without hand labels work on real photographs?

3

Number of views

Why do multi-view counterfeit models fail with fewer views, and can training fix it?

4

Unseen prints

How much does accuracy drop on counterfeit prints never seen in training?

5

Safety

Can the glass give counterfeit advice that never accuses a genuine note?

6

Bangla OCR

Which part of a Bangla OCR pipeline matters most for a wearable camera?

What existing devices miss

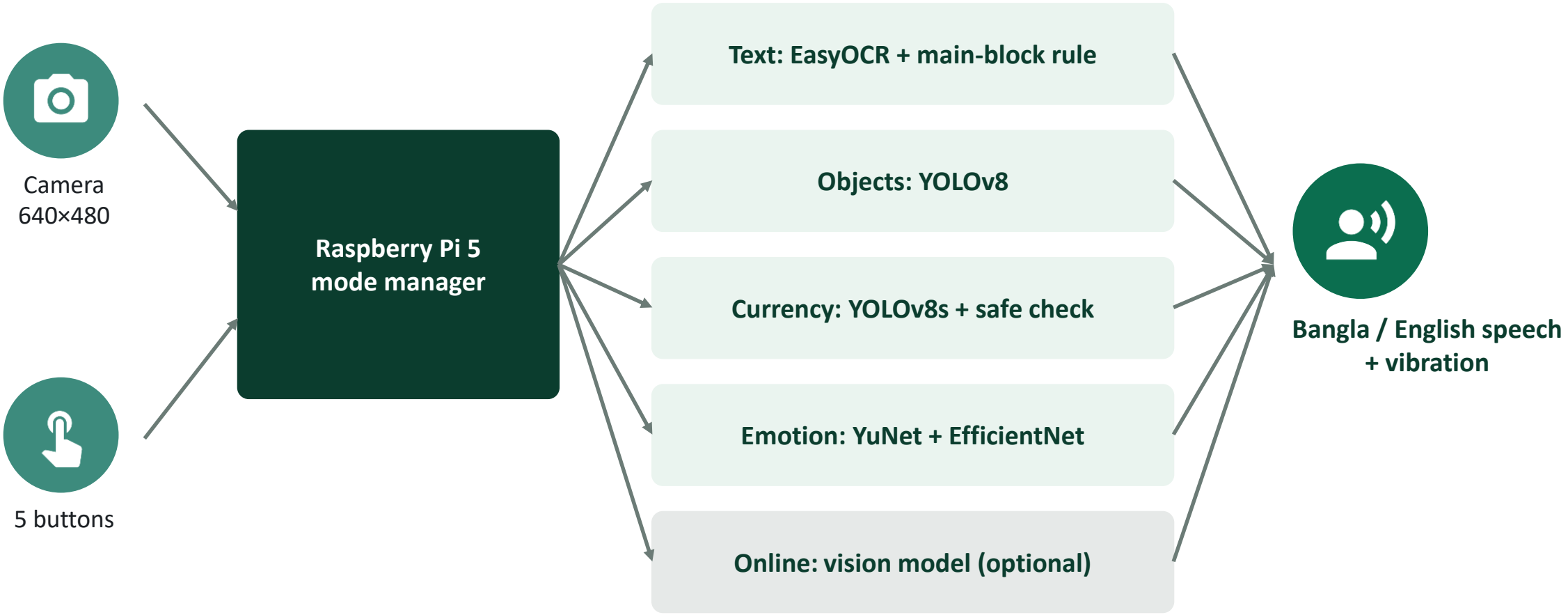
System	Reads Bangla	Knows Taka	Works offline	Wearable
OrCam MyEye	No	No	Yes	Yes
Google Glass scene analysis	No	No	No	Yes
AIris (2024)	No	No	No	Yes
Raspberry Pi reading aids	No	No	Partly	Partly
SARA (ours)	**Yes**	**Yes**	**Yes**	**Yes**

Online: the online description mode needs the internet; the other four modes do not.

The gap

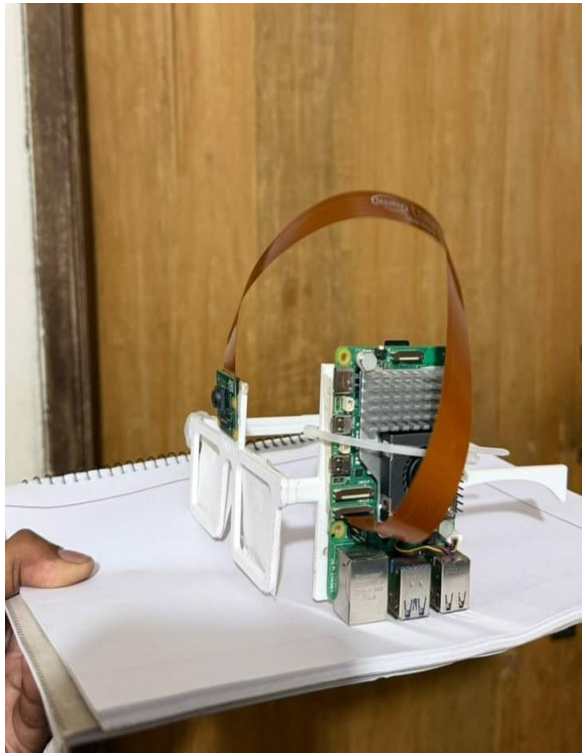
No device combines Bangla, Taka and offline use. Counterfeit studies never check whether test notes are copies of prints seen in training.

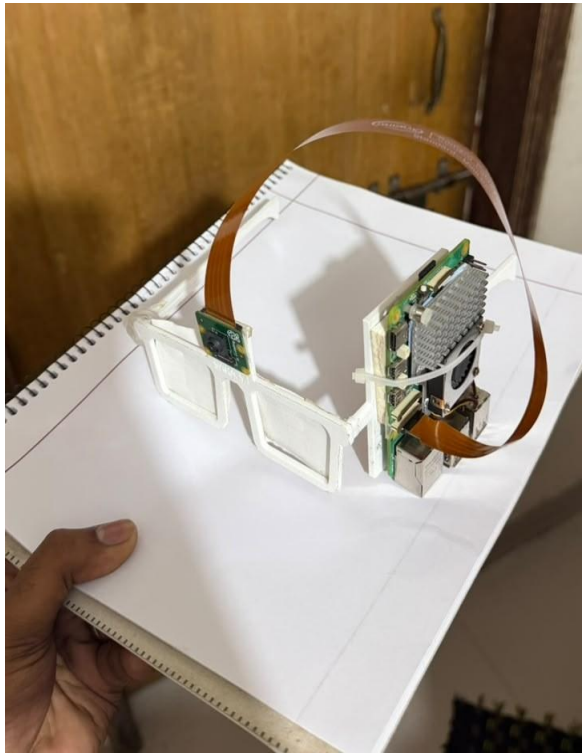
How the glass works



All models run on the Pi's CPU. Every event is written to a small CSV field log for later study.

From design to a first prototype







- Camera above the bridge of a 3D-printed frame
- Raspberry Pi 5 with fan tied to the right temple
- Ribbon cable loops from camera to board
- Next: board, buttons and battery in a pocket enclosure

The first prototype is heavy on one side and the fan sits by the ear. It is enough to test the software on a person.

Taka detection: 22,333 training images, no hand labels



5,073 note photos are pasted onto real COCO pictures with random angle, scale, light and blur. The program writes the box labels itself.

0.995

mAP@0.5 on 2,282 held-out composites

91.5%

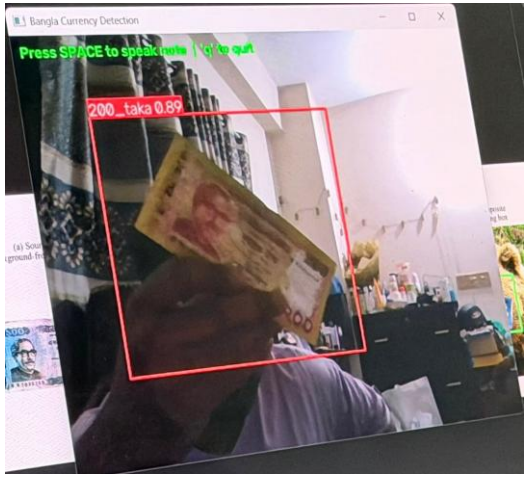
correct on 1,536 photos taken by others

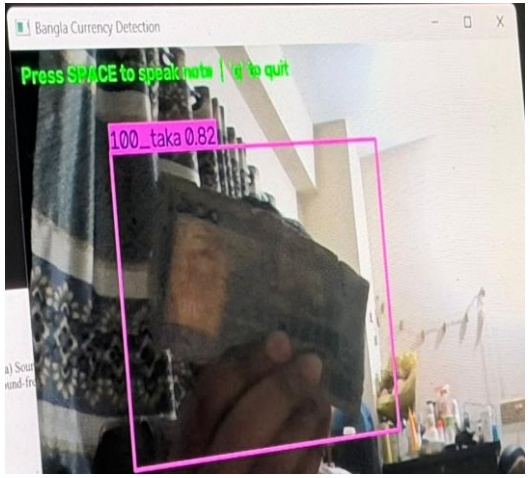
0.992

mAP@0.5 on 566 pictures with new backgrounds

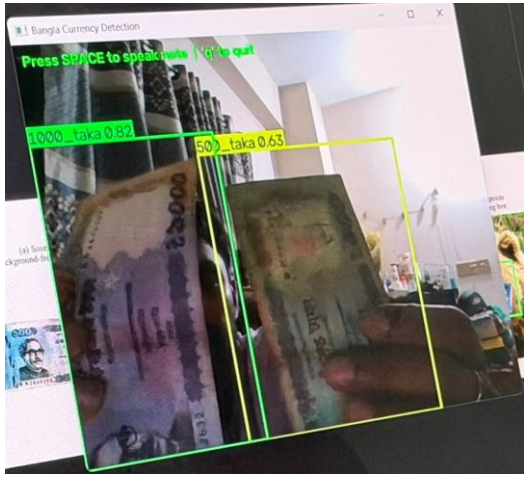
Live test with real notes

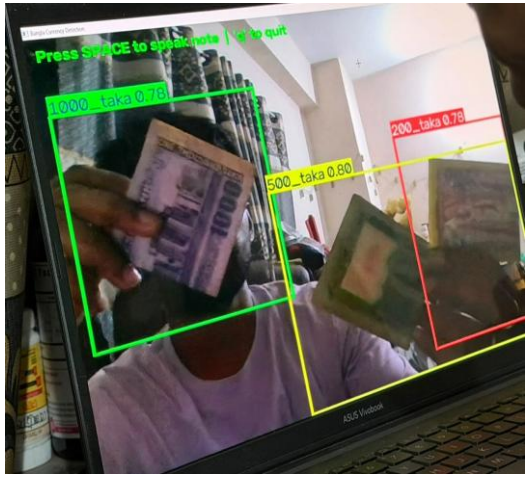








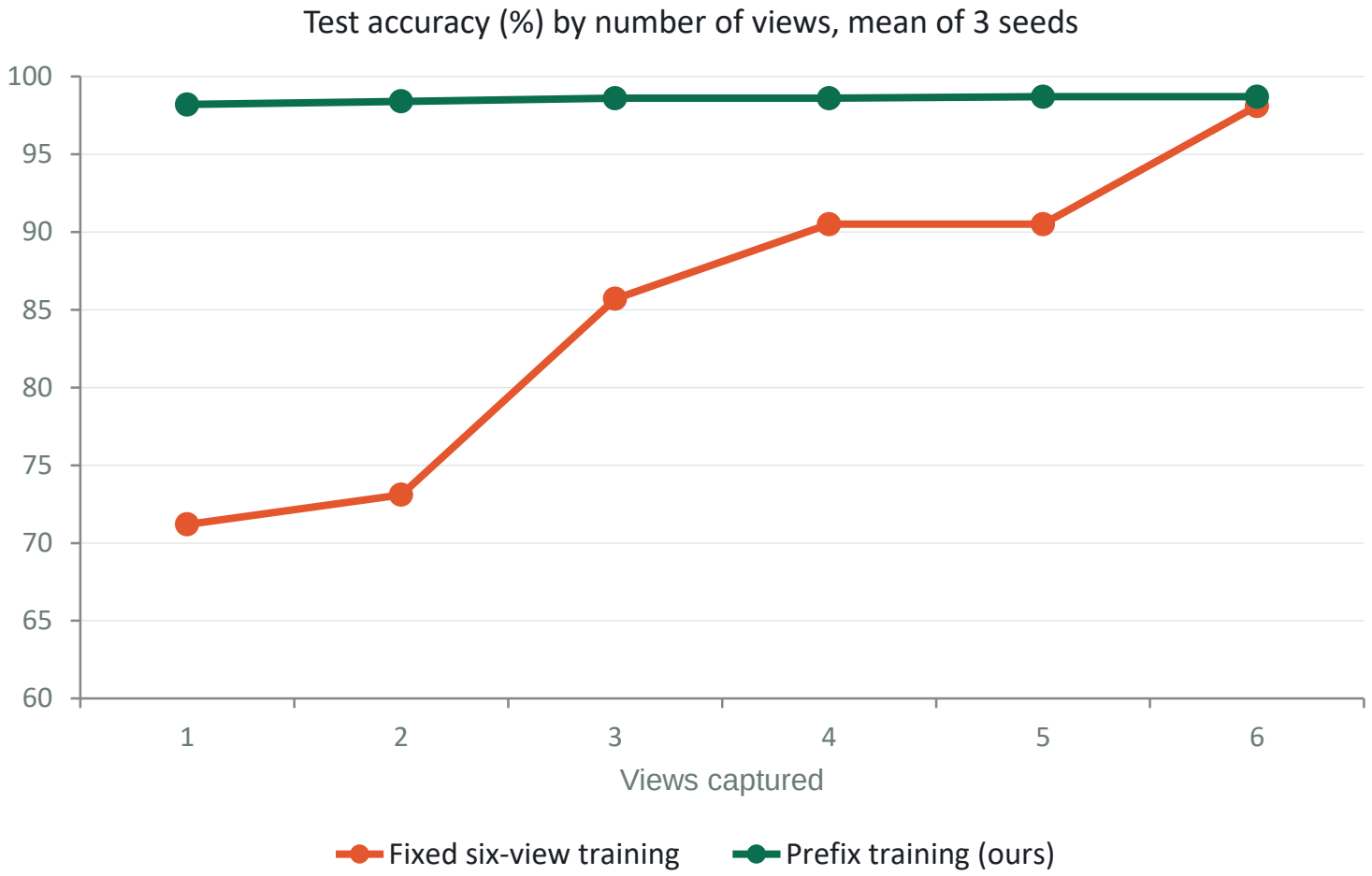




- 10, 50, 100, 200, 500 and 1,000 Taka named at 0.63–0.89 confidence
- Works on worn and folded notes and with two or three notes at once
- Boxes are loose and can overlap when notes are close

Informal sessions with a laptop webcam; no error rate was counted.

Counterfeit check: one view is enough with prefix training



71.2% → 98.2%

accuracy with a single view, JaalTaka, 3 seeds

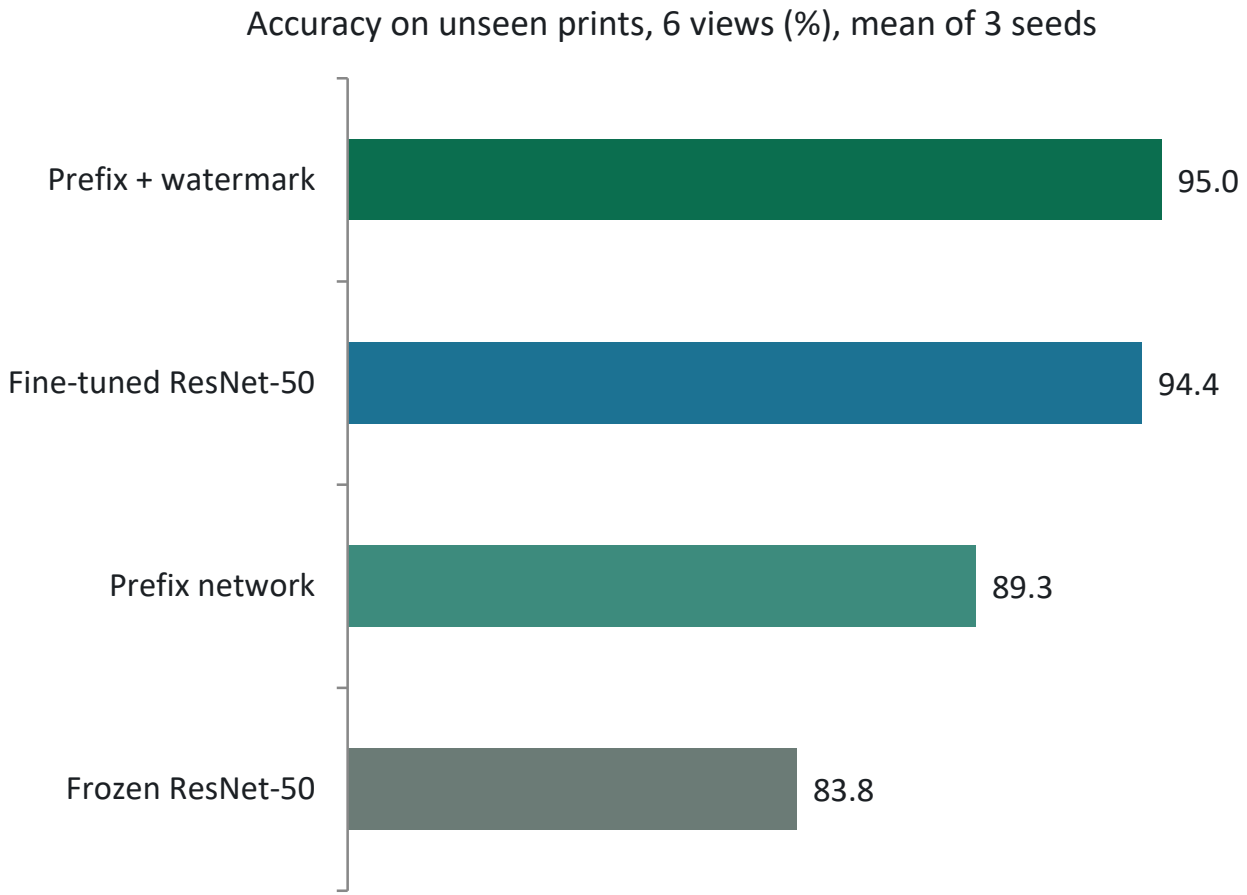
- A blind user cannot promise six views
- Training on random first-k views removes the collapse
- No loss at six views; stable across seeds

Unseen counterfeit prints: a stricter test

279 of 322

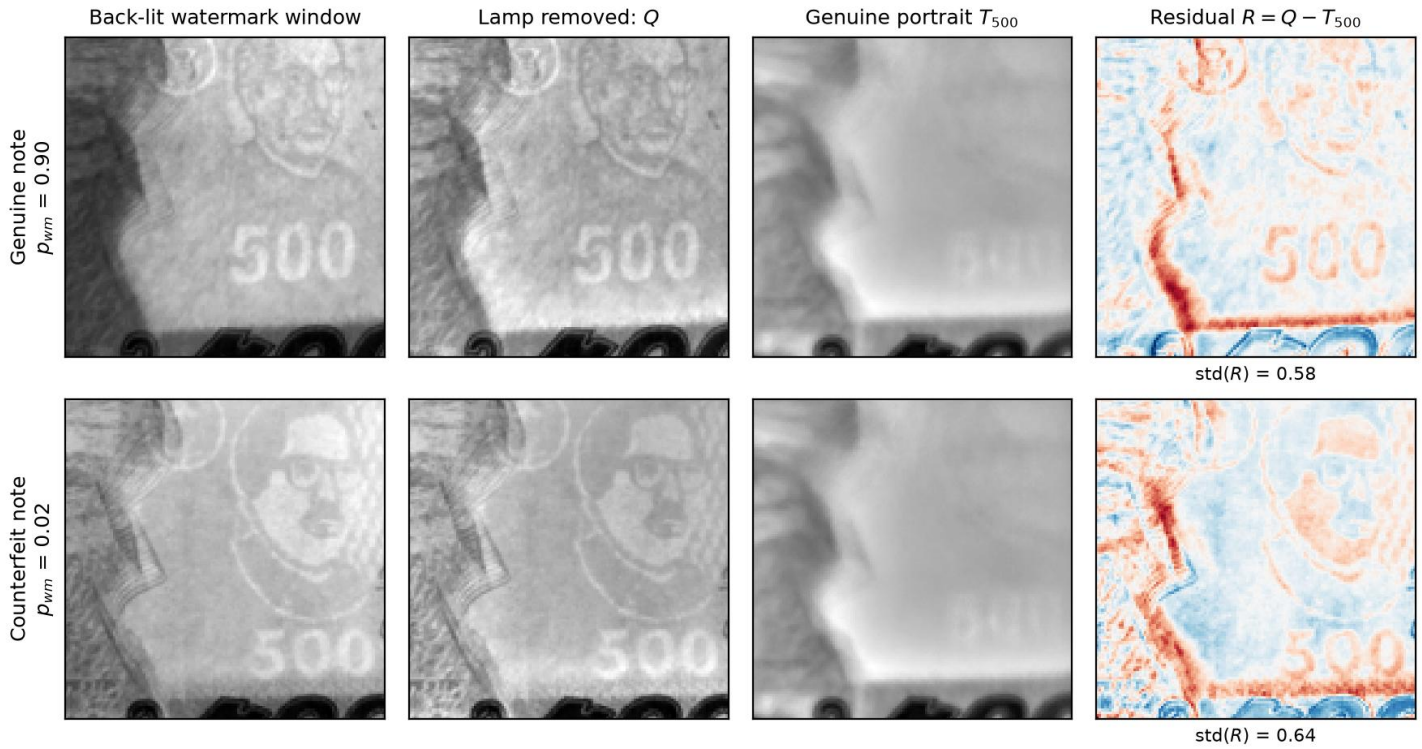
readable 500 Taka counterfeits carry the same serial number

The usual split puts copies of one print in training and test. A serial lookup alone scores 90.2%. We built a split where every test counterfeit is a new print.



Adding the watermark raises accuracy to 95%, the same on all three seeds, level with a fully fine-tuned ResNet-50.

DMWR: a tiny, physics-based watermark checker



1. Remove the lamp: Q = I / (G∗I)
2. Genuine portrait of the note value: T = median of Q over genuine training notes
3. Residual R = Q − T → two small CNN streams + 4 summary numbers → "watermark clear?"

91.5%

on unseen counterfeit prints (3 seeds, ±0.3)

47,818

parameters, 0.21 MB: 46× smaller than MobileNetV2

p ≥ 0.375

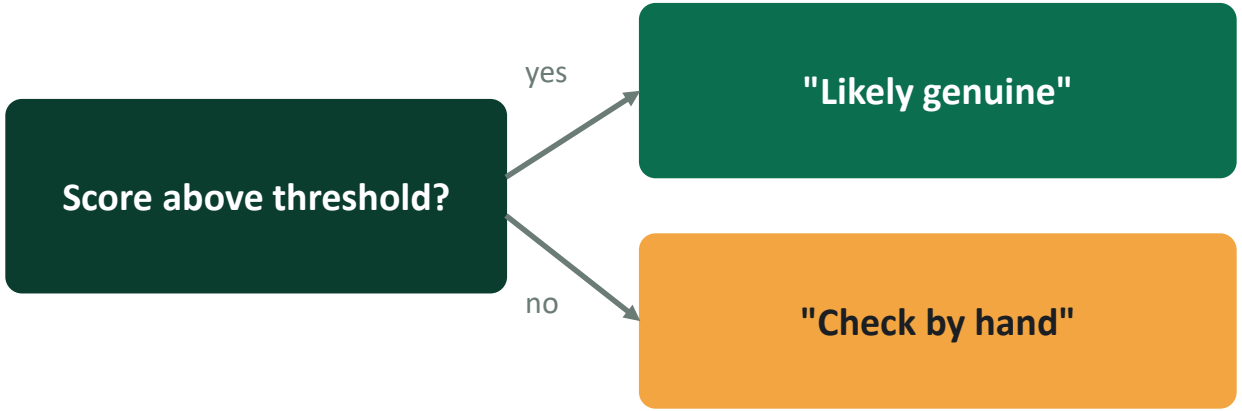
no significant difference from MobileNetV2 (92.9%)

Safe before decisive: the glass never says "counterfeit"

0 / 88

test counterfeits passed as "likely genuine"

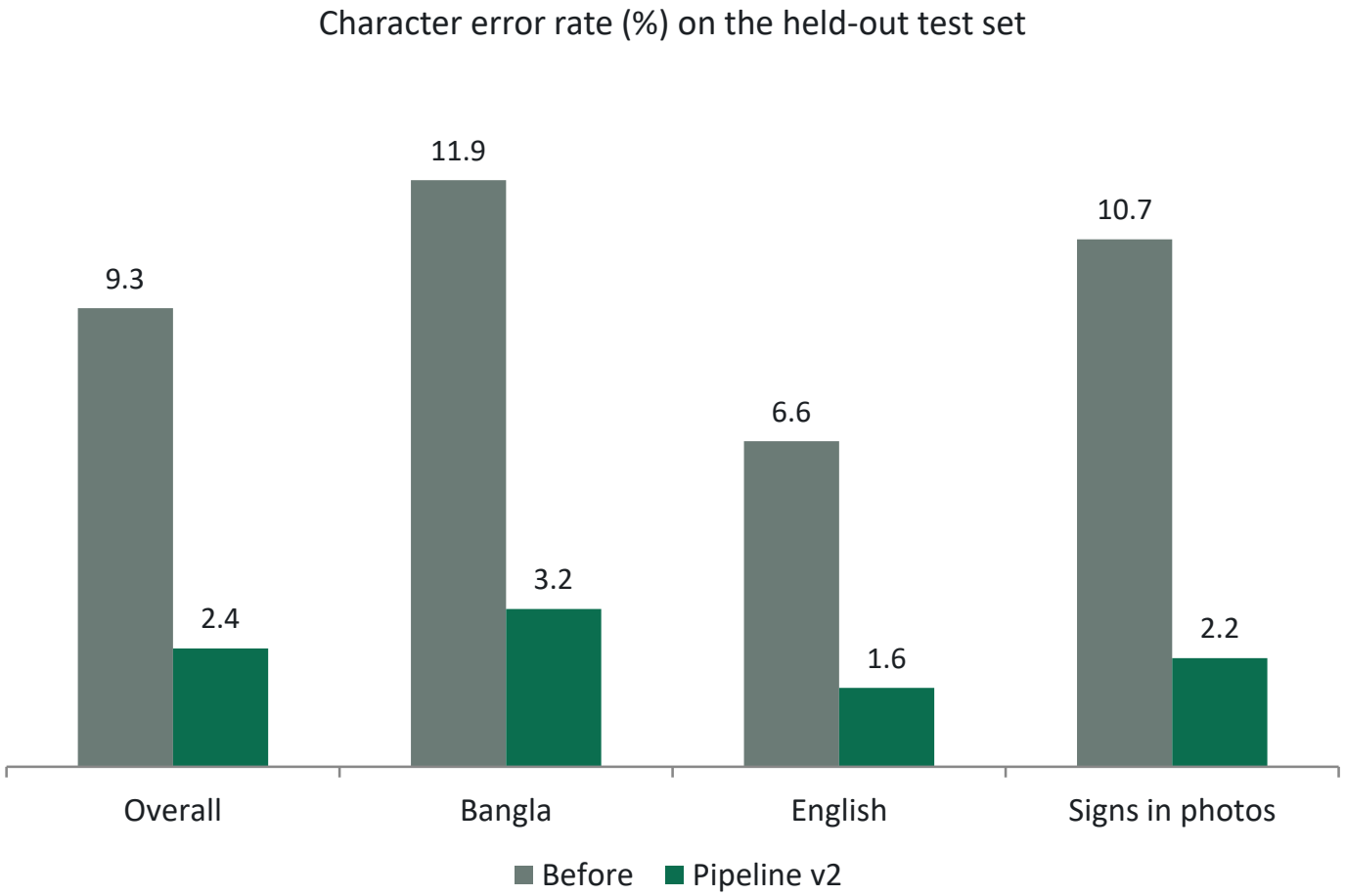
Threshold = highest score of any validation counterfeit, so a new counterfeit passes with probability 1/(n+1) = 1.1%.



When it is not sure

57.5% of genuine notes are confirmed from close-up views. Otherwise the glass asks for a quick check by hand, and the guided watermark check gives a second opinion.

Bangla reading: error cut by three quarters



What helped

- Reading only the main block of text
- Tuned detector settings, CLAHE
- Simple clean-up rules

What did not

- Binarisation, multi-scale reading
- Tesseract, TrOCR, PaddleOCR
- Dictionaries and language models

Benchmark with correctly shaped Bangla; every choice made on validation data.

Running on the Raspberry Pi 5

327 ms

currency check, median

50 ms

watermark check

207 ms

object scan

18.7 s

text reading: too slow

77.1 °C

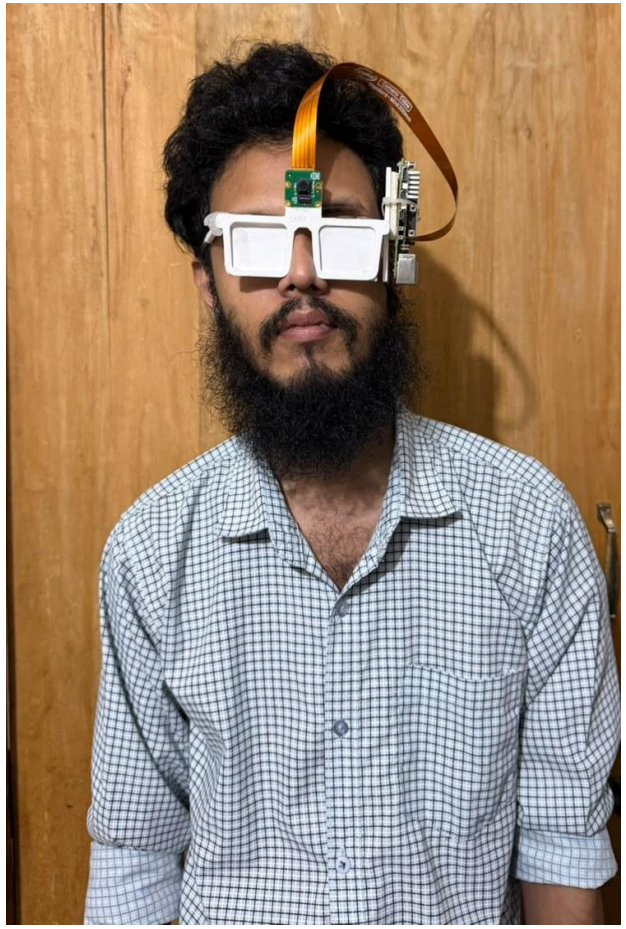
peak, no throttling in 30 min

1.48 GB

peak memory

Testing the worn glass





First field test

With the glass camera, currency was unreliable and Bangla text was not read. We suspect the 640×480 fixed-focus picture; the laptop webcam works.

Trials and survey

Sessions with a visually impaired participant have started. A Bangla voice questionnaire records answers; results will follow.

Ongoing work and next steps

1

Tune the glass camera

Higher resolution or autofocus; test set taken through the glass

2

Complete the user study

Trials started; 8–12 participants: task time, accuracy, SUS and NASA-TLX

3

Collect more prints

About 100 unseen counterfeit prints with a bank or the police

4

Faster reading

Line-by-line page reading; lighter text detector

5

Pocket enclosure

Move board, buttons and battery off the frame; measure battery

Thank you

Questions are welcome

- An offline glass that speaks Bangla, reads text, names objects and recognises Taka on a Raspberry Pi 5
- New methods: prefix training, a safe verdict, and the DMWR watermark checker
- Next: tune the glass camera and complete the user trials

